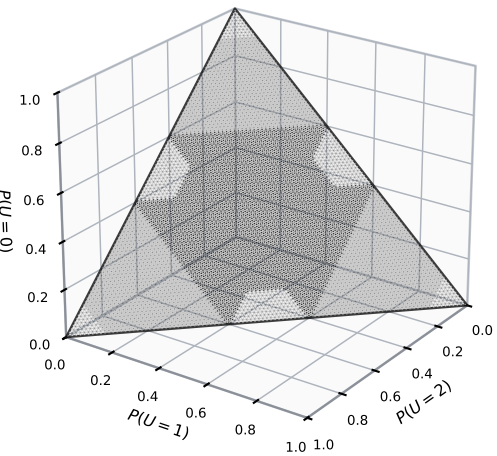
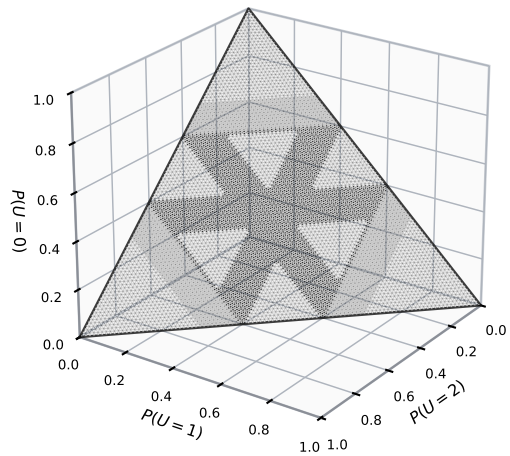


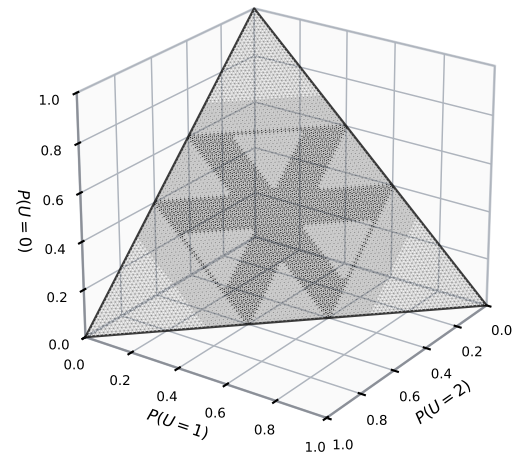
$$P(X, Y) = [(0.1, 0.3)^T, (0.3, 0.3)^T]$$



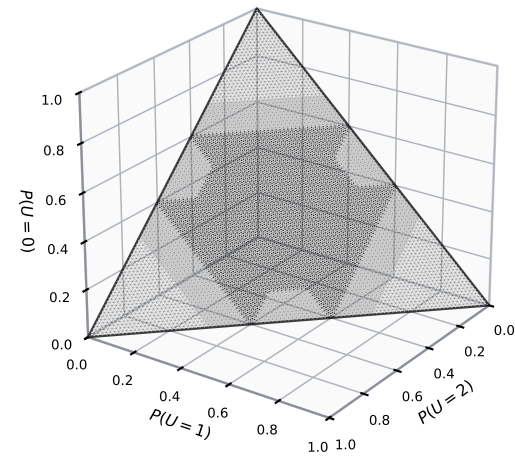
$$P(X, Y) = [(0.3, 0.1)^T, (0.3, 0.3)^T]$$



$$P(X, Y) = [(0.3, 0.3)^T, (0.1, 0.3)^T]$$



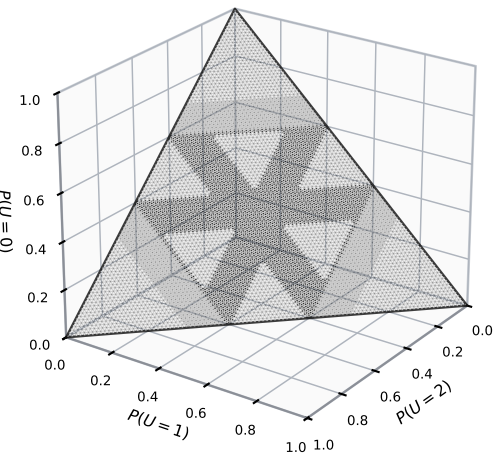
$$P(X, Y) = [(0.3, 0.3)^T, (0.3, 0.1)^T]$$



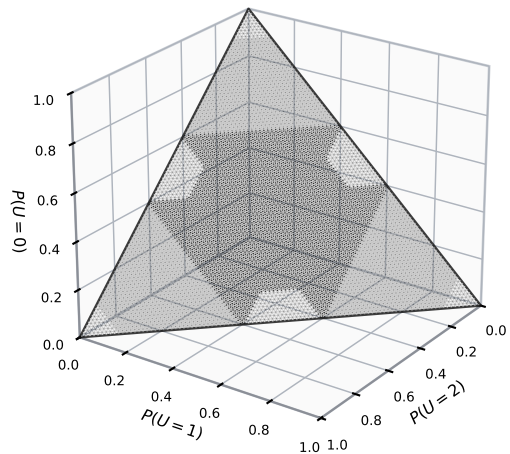
The tight identification lower bound of $P_Y(do(x))$ and ATE

$(P_L^P)^c$ (solid): non-vanilla, ATE: non-vanilla
 P_L^P/P_L^{ATE} (solid): vanilla, ATE: non-vanilla
 P_L^{ATE} (solid): vanilla, ATE: vanilla

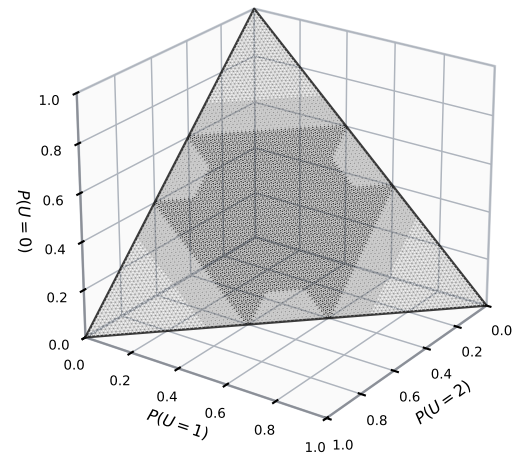
$$P(X, Y) = [(0.1, 0.3)^T, (0.3, 0.3)^T]$$



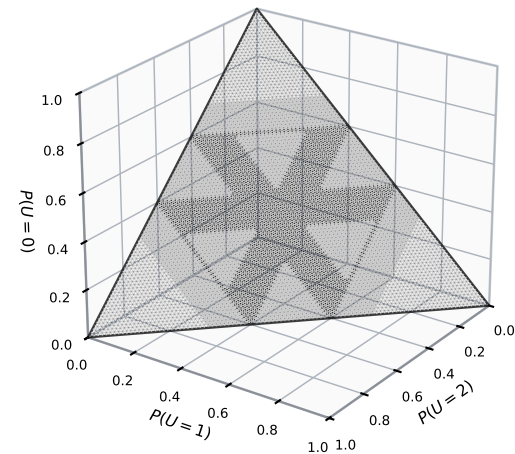
$$P(X, Y) = [(0.3, 0.1)^T, (0.3, 0.3)^T]$$



$$P(X, Y) = [(0.3, 0.3)^T, (0.1, 0.3)^T]$$



$$P(X, Y) = [(0.3, 0.3)^T, (0.3, 0.1)^T]$$



The tight identification upper bound of $P_Y(do(x))$ and ATE

$(P_U^P)^c$ (solid): non-vanilla, ATE: non-vanilla
 P_U^P/P_U^{ATE} (solid): vanilla, ATE: non-vanilla
 P_U^{ATE} (solid): vanilla, ATE: vanilla